



BIOLOGY CH.10 — EXCRETORY SYSTEM — FULL MIND MAP

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Topics in this chapter:

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1. Human Excretory System - Components

- Human excretory system: One pair kidneys, one pair ureters, single urinary bladder, single urethra. [Q619]
- Ureter: thin muscular tube (3-4mm), kidneys ko bladder se connect karti hai, bilateral hoti hain. [Q627, Q634]
- Urinary bladder: urine receive karta hai, micturition se expel hota hai. Muscular bladder nervous control mein hota hai isliye hum urinate ki urge control kar sakte hain. [Q646]
- Urethra: bladder se body exterior tak, urine nikalne ka passage. Male mein 20 cm long, urine aur sperm dono ka common passage. Female mein sirf urine carry karti hai. [Q649, Q653]
- Vas deferens urinary bladder se aane wali tube ke saath milkar urethra banta hai. [Q650]
- Prostate gland: urinary bladder ke neeche, walnut-sized, between bladder and rectum/penis. Urethra iske centre se guzarta hai. Functions: semen produce, filter, erection, UTI protection, urine flow control. [Q648, Q654]

2. Kidneys - Structure & Function

- Kidneys bean-shaped (seem ke aakar ki), abdomen mein backbone ke dono taraf. Location: last thoracic aur third lumbar vertebra ke beech. Weight: 120-170g each. [Q621]
- Kidney functions: urine produce karna, Renin hormone (BP regulate), blood filter karna, urea/uric acid remove karna, water & salt concentration maintain karna, vitamin D produce karna. [Q601, Q603, Q606]
- Nephrology = kidneys ka study. Calcium oxalate stones = sabse common kidney stones. [Q606]
- ⚠ Trap: Kidney ke filtration units ko Neurons nahi kehte - ye Nephrons hain. Neurons brain/nervous system ki cells hain. [Q621]

3. Nephron - Functional Unit

- Nephron = kidney ki functional, structural, anatomical unit. Har kidney mein ~8 lakh se zyada nephrons (1 million per kidney bhi bola jata hai). [Q617, Q621, Q623, Q647]
- Nephron components: Renal corpuscle (Glomerulus + Bowman's capsule) → PCT (Proximal Convoluted Tubule) → Loop of Henle → Distal Convoluted Tubule → Collecting duct → Renal vein. [Q630]
- Bowman's capsule: double-walled, cup-shaped, glomerulus ko surround karti hai. Function: ultra-filtration → glomerular filtrate banati hai. [Q620, Q639, Q645]
- Glomerulus: tiny blood vessels ka cluster/tuft, main filtering unit. Blood cells aur large proteins blood mein hi rehte hain. [Q620, Q639]
- Collecting duct: nephrons ko minor calyx ya renal pelvis se directly connect karti hai. [Q629]
- Long tubular structure ka advantage: excess water reabsorb karne ke liye time aur space milta hai. [Q623]
- Loop of Henle: PCT se distal convoluted tubule tak. [Q620]
- Renal tubules ki walls epithelial tissue se bani hoti hain - cuboidal epithelium. [Q613, Q625]

4. Urine Formation - 3 Stages

- Correct sequence: Filtration → Selective Reabsorption → Secretion/Elimination. [Q601, Q608]
- Filtration (Glomerular filtration/Ultrafiltration): Glomerulus blood se water aur substances filter karta hai. Filtration membrane blood cells aur large proteins ko blood mein rakhti hai. [Q608]
- Reabsorption: ~99% filtrate reabsorb hota hai. Glucose, water, salts, amino acids reabsorb hote hain. [Q602, Q615]
- Secretion: Waste ions aur H⁺ ions blood se tubule mein aate hain - acid-base balance ke liye. [Q608]
- Normal adult mein initial filtrate: ~180 L/day - isme se sirf 1-2 L urine excrete hota hai (baki reabsorb). [Q615, Q621, Q641, Q651]
- Reabsorption ki amount determine hoti hai: additional water aur dissolved waste excrete hone ki matra se. [Q615]
- GFR (Glomerular Filtration Rate) = per minute glomerulus dwara utpann filtrate ki matra. [Q641]

5. Urine Composition & Volume

- Adult human daily urine: 1000-2000 mL (1-2 L). [Q632]
- Urine composition: 95% water, 2.5% urea, 2.5% other waste products - yellowish, slightly acidic. [Q607, Q632, Q640]
- Substances jo kidneys reabsorb NAHI karti: Creatinine (muscle waste; >1.2 mg/dL women, >1.4 mg/dL men = kidney dysfunction). Glucose, amino acids, salts reabsorb hote hain. [Q628]

6. Dialysis & Hemodialysis

- Dialysis tab use hota hai jab kidneys kaam karna band kar de - waste aur excess fluid remove karta hai. Types: Hemodialysis (artificial kidney, blood bahar clean hota hai) aur Peritoneal dialysis (abdomen lining blood filter karti hai andar). [Q631, Q644, Q651]
- Hemodialysis: semipermeable tubes dialysis solution mein - waste remove hote hain, purified blood wapas pump hota hai. [Q636]
- Dialysis fluid mein blood ke samaan osmotic pressure hota hai aur nitrogenous wastes se rahat hota hai - taaki fluid imbalance na ho. Urea diffusion se remove hota hai. [Q637]
- Dialysis ka principle: Diffusion, convection, osmotic pressure. [Q626, Q636]
- Artificial kidney nitrogenous waste remove karta hai - is process ko Haemodialysis kehte hain. [Q631]

7. Excretion in Plants

- Plants mein defined excretory system NAHI hoti kyunki ye vegetative parts se excrete karte hain. [Q635]
- Plant excretion methods: Transpiration (excess water), Diffusion (gas exchange), Storage (vacuoles, old xylem, shed leaves). [Q605]
- ⚠ Trap: Plants ke paas waste removal ke liye specialized organs NAHI hote. [Q609]
- Plant waste products: O₂, CO₂, water vapour (gaseous), Raphides, tannins, resins, gum, rubber, essential oils (solid). [Q616, Q635]
- Gaseous wastes stomata aur lenticels se remove hote hain. Solid wastes leaves/bark mein store hote hain jo shed hone par eliminate hote hain. [Q616, Q635]
- ⚠ Trap: Photosynthesis ka waste product Oxygen hai, starch NAHI. [Q616]
- Plants very slowly aur very little waste excrete karte hain. Respiration ka CO₂ aur photosynthesis ka O₂ ek doosre ke raw material hote hain. [Q609]
- ⚠ Trap: Plants sweating se waste eliminate NAHI karte - ye animal method hai. [Q624]

8. Types of Nitrogenous Waste

- Three main types: - Ammonia (Ammonotelic): Most toxic. Aquatic animals (fish, protozoans) - abundant water dilute kar sakti hai. [Q607, Q610]
- Human urine: 95% water, 2% urea, 0.1% creatinine, 0.03% uric acid. [Q612]
- Cow urine: 95% water, 2.5% urea, minerals, 24 types salts. [Q612]
- Lactic acid nitrogenous waste NAHI hai - ye anaerobic respiration ka by-product hai muscles mein. [Q638]
- Animal excretory products: Ammonia, CO₂, Urea, Uric Acid, Guanine, Creatine. Modes: Ammonotelism, Ureotelism, Uricotelism, Aminotelism, Guanotelism. [Q638]
- Ammonia (Ammonotelic): Most toxic. Aquatic animals (fish, protozoans) - abundant water dilute kar sakti hai. [Q607, Q610]
- Urea (Ureotelic): Medium toxicity. Humans, cows, mammals. Liver mein urea cycle (ornithine cycle) se banta hai. [Q607, Q611, Q612]

- Uric acid (Uricotelic): Least toxic, minimum water chahiye (1g = 10ml water). Reptiles, birds, insects. Liver mein 'Inosinic pathway' se banta hai. [Q607, Q612, Q643]

9. Other Organs in Excretion

- Excretory organs humans mein: Skin, Liver, Large intestine, Lungs, Kidneys. Small intestine excretory organ NAHI hai. [Q604]
- Lungs: gas exchange (CO₂ remove). Skin: sweat (water, salts, urea), temperature regulation. Large intestine: water absorb, feces move karna. Liver: urea produce karna. [Q604]
- Cartilage tissue kidney mein absent hoti hai - nose, ear, larynx mein hoti hai. [Q622]

10. Medical Techniques & Miscellaneous

- Ultrasonography (Ultrasound): kidney/gall bladder stones aur pregnancy mein fetus examine karne ke liye - non-invasive, high-frequency sound waves. [Q642]
- ECG = heart rhythm. MRI = 3D anatomical images. Echocardiography = heart imaging. [Q642]
- Ultrafiltration excretion se related hai - kidney ke glomerulus mein hoti hai. [Q652]
- Respiration = C-C bonds break by oxidation. Digestion = complex food to simple absorbable forms. [Q652]

11. Missing Facts — Plant Excretion, Alimentary Canal & Diagram

- Plants ke excretory products mein 'Carbon Dioxide' (CO₂) inhale karne par humans ke liye harmful hai — Gums medicines-cosmetics mein use hoti hain, Resins adhesives-food glazing mein, Oxygen essential hai. [Q614]
- Humans mein alimentary canal ka 'Anus' part waste material excrete karta hai — Large intestine indigestible matter process karke water absorb karta hai, feces rectum-anus se expel hoti hai. [Q618]
- Excretory system diagram mein ureters aur urinary bladder ki sahi position 'Diagram 1' mein dikhti hai — Liver se urea blood mein jaata hai, kidneys filter karte hain, urine ureters ke through bladder mein jaata hai, urethra se expel hoti hai. [Q633]